

Problem Set 2

Topics: Dimensionality Reduction, Logistic Regression, Support Vector Machine, Decision Tree, Ensemble Learning

Due date: 27.07.2026, 23:59

Grading: 40% of your final grade

Notes:

- **Report submission:** Submit your answers/reports to **Moodle** before the deadline. Submissions via email will NOT be accepted. Only the reports submitted to Moodle will be evaluated.
- **Report format:** You need to submit a zip file containing: (1) a report as a PDF document that contains your solutions to all problems, and (2) a supplementary Jupyter notebook (.ipynb) or similar code files.

The zip file should be named as “**Matriculation number, Name**”. For example, “**03709721, Jane Doe.zip**”. This information should also be provided at the beginning of the report and the notebook/code.

Note that the notebook/code should be submitted with the outputs (i.e., after running each cell). You can use any programming language you like.

- **Report evaluation:** You are evaluated based on your reports and the final answers. So, you have to describe and explain the steps to solve each problem as clearly as possible.
- ⚠ You need to solve these problems individually, and all text needs to be written in your own words. Solutions will be checked for plagiarism and originality. Copying others’ solutions will NOT be tolerated and will result in failing this course.
- ⚠ Late submission will NOT be accepted. If you encounter any technical issues, please contact the corresponding lecturer.
- ⚠ Handwritten answers are NOT accepted. All submissions must be typed digitally.

1. Problem 1 (35 points)

We have a dataset from a survey which records travel mode for a sample of travellers in a given city. We generate two binomial logistic regression models with travel mode as the dependent variable: 0 refers to walking and 1 refers to travel by car.

The results of the two models are given below. In Model 1, the independent variables are travel time in seconds (**TrvlTime**), traveller age in years (**Age**), the number of times the individual travels

during the survey period (Indiv_trvl_frequency), the number of vehicles owned by the household (HHVeh), and the number of people in the household (HHsize). Model 2 contains the same variables as Model 1 minus the variable Indiv_trvl_frequency.

Model 1 Results:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.126e+00	2.700e-02	-78.723	< 2e-16 ***
TrvlTime	9.786e-04	1.037e-05	94.334	< 2e-16 ***
Age	1.716e-02	3.858e-04	44.485	< 2e-16 ***
Indiv_trvl_frequency	1.611e-02	4.748e-04	33.937	< 2e-16 ***
HHVeh	1.012e+00	8.924e-03	113.359	< 2e-16 ***
HHsize	-2.463e-02	3.285e-03	-7.497	6.54e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

n = 244493

BIC = 95815.66

Model 2 Results:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.038e+00	2.689e-02	-75.778	< 2e-16 ***
TrvlTime	9.465e-04	1.026e-05	92.282	< 2e-16 ***
Age	1.711e-02	3.876e-04	44.149	< 2e-16 ***
HHVeh	1.188e+00	7.544e-03	157.549	< 2e-16 ***
HHsize	-9.367e-03	3.244e-03	-2.888	0.00388 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

n = 244493

BIC = 97047.6

Please include your **code** for Problem 1.3. Complete Jupyter Notebooks (*.ipynb*) or equivalent script outputs must be submitted alongside the report.

1. Discuss the performance of both models in terms of overall model performance and the parameter results. Which model performs best for classifying the travel mode? (10 points)
2. We have two sets of input values from two survey respondents for the independent variables as given in Table 1. Based on your identification of the best model above, calculate the probability of these two respondents choosing the car mode, and comment on your results. (15 points)

Table 1: Inputs from two survey respondents

Respondent	TrvlTime	Age	Indiv_trvl_frequency	HHVeh	HHsize
A	3650	60	24	2	2
B	200	26	7	0	1

3. We have received a sample of 5000 observations from the survey data, which is given in the file `mode_choice_pcasample.csv`. Undertake a principal components analysis of the independent variables. Nominate an appropriate number of principal components to represent variation in the data, and comment on your results. Please include your coding (or appropriate working out in any type of software). (10 points)

2. Problem 2 (30 points)

Consider a binary classification problem with two-dimensional feature space. You are given a dataset containing eight samples (see Table 2). The plots in this problem should be created digitally rather than hand-drawn. You are free to use any software or programming language you like.

Table 2: Binary classification dataset

x1	x2	y
1.5	1.0	1
1.5	3.0	1
3.5	1.0	1
3.5	3.0	1
0.5	0.0	-1
0.5	2.0	-1
2.5	0.0	-1
2.5	2.0	-1

*The plots in this problem should be created **digitally** rather than hand-drawn. You are free to use any software or programming language you like.*

1. Show the given data points on a plot, with the two classes labelled by different markers. (5 points)
2. Suppose we want to build a simple decision tree with the following if-else decision rule:

$$\hat{y} = \begin{cases} 1, & \text{if } x_1 \geq 1, \\ -1, & \text{if } x_1 < 1. \end{cases}$$

Compute the information gain rate from this decision rule. (5 points)

3. Train a soft margin linear SVM model on the dataset. Show the decision boundary and label support vectors on the plot. (5 points)
4. Find an explicit mapping in the two-dimensional space to transform the linearly non-separable dataset into a linearly separable one. Show the transformed data points on a plot. (6 points)
5. Make predictions on four new samples $(-0.5, 1.0)$, $(-0.5, 3.0)$, $(0.5, 4.0)$, and $(2.5, 4.0)$ based on your mapping. (4 points)
6. Apply the kernel trick on the original dataset. Show the kernel matrix of all the points on a plot. (You don't have to train the nonlinear SVM.) (5 points)

3. Problem 3 (35 points)

Topic: Decision trees, bootstrap aggregation, random forests, gradient boosting, and model evaluation.

Context: A public transport control centre wants to predict whether an operational incident will become a *severe delay*. Throughout this problem, let

$$Y = 1 \quad \text{mean severe delay,} \quad Y = 0 \quad \text{mean non-severe delay.}$$

Please include your code for Problem 3.4. Complete Jupyter Notebooks (.ipynb) or equivalent script outputs must be submitted alongside the report.

3.1 Decision-tree splitting criteria (12 points)

At one node of a decision tree, there are 12 historical incidents:

$$7 \text{ severe,} \quad 5 \text{ non-severe.}$$

Two candidate binary splits are considered:

Candidate split	Left child	Right child
A: <code>current_delay_min</code> ≥ 10	5 severe, 1 non-severe	2 severe, 4 non-severe
B: <code>major_incident</code> = yes	5 severe, 0 non-severe	2 severe, 5 non-severe

Use the following definitions:

$$H(S) = - \sum_k p_k \log_2(p_k), \quad G(S) = 1 - \sum_k p_k^2.$$

1. Compute the entropy and Gini impurity of the parent node. (3 points)
2. Compute the weighted child entropy and information gain for Split A and Split B. (3 points)
3. Compute the weighted child Gini impurity and Gini reduction for Split A and Split B. (3 points)
4. Which split should be selected first? Explain in one or two sentences in the transportation context. (3 point)

3.2 Bootstrap sampling and out-of-bag observations (9 points)

A random forest is trained on a dataset with n observations. Each tree is trained on a bootstrap sample of size n , sampled *with replacement* from the original dataset.

1. For a fixed observation i , derive the probability that i is not selected in one bootstrap sample. Also, give the probability that i is selected at least once. (3 points)
2. Show the limiting values of these two probabilities as $n \rightarrow \infty$. Interpret the result in words. (2 points)
3. For $n = 5000$, compute the expected number of unique observations appearing at least once in a bootstrap sample and the expected number of out-of-bag observations. (2 points)
4. A small dataset contains five distinct incidents $\{A, B, C, D, E\}$. A bootstrap sample of size 5 is drawn. How many ordered bootstrap samples are possible? How many unordered bootstrap datasets are possible if only the multiplicities of A, B, C, D, E matter? (2 points)

3.3 Ensemble voting and boosting intuition (4 points)

1. Suppose five independent weak classifiers each predict the correct class with probability $p = 0.65$. The ensemble predicts by majority vote. Compute the probability that the ensemble prediction is correct. (2 points)
2. In a gradient boosting regression example, the observed delays are

$$y = [5, 9, 14, 18]$$

and the current model predicts

$$\hat{y}^{(0)} = [8, 8, 8, 8].$$

Compute the residuals $r_i = y_i - \hat{y}_i^{(0)}$. A new small tree predicts corrections

$$h_1(x) = [-2, 1, 5, 5].$$

With learning rate $\eta = 0.2$, compute the updated predictions

$$\hat{y}^{(1)} = \hat{y}^{(0)} + \eta h_1(x).$$

(2 points)

3.4 Coding exercise: severe-delay prediction (10 points)

In this part, you will compare a decision tree, a random forest, and a gradient boosting classifier on a synthetic but transportation-motivated dataset. Generate the dataset using the following Python code. If you use another programming language, generate an equivalent dataset using the same variables and equations.

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(42)
n = 1200

df = pd.DataFrame({
    "peak_hour": rng.binomial(1, 0.45, n),
    "rain": rng.binomial(1, 0.30, n),
    "major_incident": rng.binomial(1, 0.12, n),
    "critical_hub": rng.binomial(1, 0.25, n),
    "current_delay_min": np.clip(rng.normal(5, 4, n), 0, None),
    "passenger_load": rng.uniform(0.2, 1.2, n),
})

score = (
    1.0 * (df["current_delay_min"] > 8)
    + 1.0 * (df["peak_hour"] & df["rain"])
    + 1.5 * df["major_incident"]
    + 0.9 * (df["critical_hub"] & (df["passenger_load"] > 0.8))
    + 0.8 * (df["rain"] & (df["current_delay_min"] > 5))
    + 0.6 * (df["peak_hour"] & (df["passenger_load"] > 0.9))
    + rng.normal(0, 0.7, n)
)

df["severe_delay"] = (score > 1.5).astype(int)
```

Use a stratified train-test split with 70% training data and 30% test data, and use `random_state=42` wherever applicable.

1. Report the class balance of `severe_delay`. Why is class balance relevant for transportation applications? (1 point)
2. Train a decision tree classifier with `max_depth=3`. Report the first split of the tree and evaluate the model on the test set using accuracy, F1-score, ROC-AUC, and the confusion matrix. (1 point)
3. Train a random forest classifier with `n_estimators=300` and `min_samples_leaf=5`. Train a gradient boosting classifier with `n_estimators=150`, `learning_rate=0.05`, and `max_depth=3`. Compare the three models using the same test metrics. (4 points)
4. Report feature importances for the three models. Which variables are most influential? Are the importances identical across methods? (2 points)
5. Based on your results, which model would you recommend as a decision-support tool for a transport control centre? Discuss the trade-off between false positives and false negatives. (2 points)